

5.4.9

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Question

Using elementary transformations, find the inverse of the following matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Solution

We know that,

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (1)$$

where $\mathbf{I}$ is a 2 X 2 identity matrix.

The following is the augmented matrix we get,

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \quad (2)$$

Solution

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow[R_2 \rightarrow \frac{R_2}{-2}]{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \quad (3)$$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \quad (4)$$

$$\text{Therefore, } \mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}. \quad (5)$$

```
#include <stdio.h>

int inverse(double* mat, double* result) {
    double a = mat[0], b = mat[1], c = mat[2], d = mat[3];
    double det = a*d - b*c;
    if(det == 0) return 0;

    result[0] = d/det;
    result[1] = -b/det;
    result[2] = -c/det;
    result[3] = a/det;
    return 1;
}
```

```
import ctypes

lib = ctypes.CDLL('./libinverse.so')
lib.inverse.argtypes = [ctypes.POINTER(ctypes.c_double),
                        ctypes.POINTER(ctypes.c_double)]
lib.inverse.restype = ctypes.c_int

# Input each element individually
a11 = float(input(Enter mat[1][1]: ))
a12 = float(input(Enter mat[1][2]: ))
a21 = float(input(Enter mat[2][1]: ))
a22 = float(input(Enter mat[2][2]: ))
```

```
# Create ctypes array row-wise
mat_array = (ctypes.c_double * 4)(a11, a12, a21, a22)
res_array = (ctypes.c_double * 4)()

# Call C function
status = lib.inverse(mat_array, res_array)

if status == 0:
    print(Matrix is singular, inverse does not exist.)
else:
    inv_matrix = [[res_array[0], res_array[1]],
                  [res_array[2], res_array[3]]]
    print(Inverse matrix:)
    for row in inv_matrix:
        print(row)
```